

CH2 455 Homework 8

Problem 1:

$$r = 3.25 \cdot 10^{-7} \text{ cm}$$

$$\rho = 5.82 \text{ g/cm}^3$$

$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (3.25 \cdot 10^{-7})^3 \approx 1.44 \cdot 10^{-19} \text{ cm}^3$$

$$\text{Mass} = \rho \cdot V = (5.82) (1.44 \cdot 10^{-19}) = \boxed{8.38 \cdot 10^{-19} \text{ g}}$$

Problem 2:

$$n = \frac{(1.2 \cdot 10^{-14}) \div 1000}{28}$$

$$\frac{[\text{g}]}{[\frac{\text{g}}{\text{mol}}]} = \boxed{4.29 \cdot 10^{-20} \text{ mol}}$$

$$(4.29 \cdot 10^{-20} \text{ mol}) \cdot (6.023 \cdot 10^{23} \frac{\text{atoms}}{\text{mol}}) = \boxed{2.58 \cdot 10^4 \text{ atoms}}$$

$$\text{Atomic \#} = 14 \frac{\text{electrons}}{\text{atom}}$$

$$(14) (2.58 \cdot 10^4) = \boxed{3.61 \cdot 10^5 \text{ electrons}}$$

$$\frac{e}{\text{atom}}, \text{ atoms} = e$$

Problem 3

$$r = 3.5 \cdot 10^{-7} \text{ cm}$$

$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (3.5 \cdot 10^{-7})^3 = 1.8 \cdot 10^{-19} \text{ cm}^3$$

$$V = 1.8 \cdot 10^{-19} \text{ cm}^3$$

Problem 4

$$\text{Mass} = \rho \cdot V = 4.82 \frac{\text{g}}{\text{cm}^3} \cdot 4 \cdot 10^{-21} \text{ cm}^3$$

$$\text{Mass} = 1.93 \cdot 10^{-20} \text{ g}$$

Problem 5:

$$n = \frac{1.93 \cdot 10^{-20} \text{ g}}{191.37 \text{ g/mol}} = 1.01 \cdot 10^{-22} \text{ mol}$$

$$1.01 \cdot 10^{-22} \text{ mol} \cdot 6.022 \cdot 10^{23} \frac{\text{atoms}}{\text{mol}} = 61 \text{ atoms}$$